MOSAIC: Certifiable Stochastic Representation Release under Structured Deployment Shift

Anonymous submission

Abstract

Deploying an edited representation requires more than showing that one probe fails on one validation split: the released representation may face an arbitrary downstream attacker and a shifted deployment population. We present MOSAIC, Minimax-Optimized Source-Agnostic Invariant Channels, a software-only method that jointly selects a stochastic representation-release channel and a task decoder while certifying privacy and utility under a structured pre-release shift. A single multinomial confidence event over fine-token laws simultaneously covers the continuum of channels selected on that same table, so the certificate incurs no channel-grid correction within that table. We derive an exact multiclass attacker envelope, an exact structured-shift theorem combining common drift with bounded differential contamination, a matching utility certificate, and a no-free-lunch result for unrestricted differential shift. For a finite transform polytope, decoder enumeration and linear optimization find the global solution. In hash-locked synthetic studies, naive continuum selection deploys contract-violating releases on 42.1% of 1,000 primary tables while MOSAIC records none. On 1,000 paired low-data tables, the transform-exact certificate safely deploys on 53.0%, versus 0.3% for its capacity-transfer fallback; at the next sample size the rates are 99.9% and 57.8%. In a fresh paired study of 325 official-method rows on five datasets, the exact objective is never worse and improves 135 rows, adding one diagnostically safe Waterbirds release. All six externally estimable selections pass their natural diagnostic. Independent programs replay 64,000 broad-comparison receipts, 10,000 exact-refinement receipts, and 650 real-feature optima with zero mismatch.

1 Introduction

Representation editing is often evaluated by fitting a probe to a frozen embedding and checking whether a designated source attribute remains predictable (Ravfogel et al. 2020, 2022a,b; Belrose et al. 2023; Jourdan et al. 2023; Avitan, Goldberg, and Elazar 2026). Probe design itself can substantially change the apparent conclusion (Hewitt and Liang 2019; Pimentel et al. 2020; Voita and Titov 2020). That evaluation leaves two deployment gaps. First, failure of a particular probe does not prevent a different downstream rule from recovering the attribute. Second, an edit chosen on a validation table can exploit sampling noise, while the deployed population can change after certification. These gaps are especially consequential when the released representation is a public interface: the designer controls the release mechanism but not the future attacker.

This paper studies a finite but expressive release model. A tokenizer fixed without the certification fold maps an internal representation to a fine token $C \in [K]$. A source-blind row-stochastic channel $M \in \mathrm{Stoch}(K, L)$ releases the only public token Z. For each task label, privacy is measured by the balanced accuracy of the Bayes-optimal source attacker on Z, not by a chosen probe family. The deployment shift occurs before the software-enforced channel and decomposes into a common Markov transformation plus a bounded fraction of source-specific contamination. This model permits large source-blind drift while explicitly charging the component that can create new source differences.

MOSAIC, or Minimax-Optimized Source-Agnostic Invariant Channels, turns this model into a certifying release rule. Its first ingredient is an exact robust attacker envelope built from simultaneous multinomial confidence regions over the fine-token table. Because the confidence event is stated before applying M, it covers every stochastic channel at once, including one optimized on the same data. Its second ingredient exactly maximizes that confidence envelope over a registered common-transform polytope, bounded differential contamination, and all residual laws; channel capacity supplies both the unrestricted boundary and a scalable fallback. The same event certifies a jointly selected task decoder. Finally, the piecewise-linear certificate is optimized globally for finite output and decoder alphabets, and the method abstains when no release satisfies the registered contract.

The contribution is not the individual use of randomized channels, distributional confidence regions, contraction coefficients, or accept-or-abstain decisions, all of which have substantial prior literatures. The technical contributions are:

- an exact finite-sample envelope for an arbitrary downstream attacker that simultaneously covers every same-table selected stochastic channel;
- an exact certificate for a common-transform polytope plus arbitrary differential contamination, with matching utility, a capacity-transfer fallback, and impossibility results at

the unrestricted and missing-source boundaries;

- a globally solved finite-alphabet release rule that jointly chooses a randomized channel and task decoder and otherwise abstains; and
- hash-locked paired experiments against plug-in selection, held-out certification, finite LTT, deterministic release, official erasure methods, a shift-unaware ablation, and an exact population oracle.

2 Related Work

Erasure and universal representations. Linear and nonlinear concept-erasure methods remove directions, covariance, or components associated with a source variable (Ravfogel et al. 2020, 2022a,b; Belrose et al. 2023; Jourdan et al. 2023; Holstege, Ravfogel, and Wouters 2025; Akbari, Afshari, and Boddeti 2025; Avitan, Goldberg, and Elazar 2026); causal interventions and leakage audits also show why a weak probe is not a universal certificate (Elazar et al. 2021; Chowdhury et al. 2025; Zarlenga et al. 2025). FARE certifies every downstream classifier for a restricted finite representation, while smooth-representation and adversarial-inference work develops other finite-sample, high-confidence fairness guarantees (Jovanovic et al. 2023; Gitiaux and Rangwala 2021; Deka and Sutherland 2023; Luo et al. 2025). MOSAIC differs narrowly: its confidence event is placed on pre-release token laws, uniformly covering a stochastic post-channel optimized on that same table and then transferred through an explicit shift class.

Randomized mappings and risk control. Probabilistic preprocessing, privacy funnels, and fair Pareto optimization already establish that stochastic channels can improve population privacy–utility tradeoffs (Calmon et al. 2017; Makhdoumi et al. 2014; Agarwal and Deshpande 2024; Kozdoba, Perets, and Mannor 2024). MOSAIC does not claim randomization as new. Risk-controlling prediction, Learn-Then-Test (LTT), conformal risk control, and Pareto testing certify registered decisions and abstain when none passes (Bates et al. 2021; Angelopoulos et al. 2025, 2024; Laufer-Goldshtein et al. 2023; Zollo et al. 2024). Robust conformal variants further address specified covariate or likelihood-ratio shifts (Tibshirani et al. 2019; Cauchois et al. 2024; Almeida et al. 2025; Zecchin et al. 2025). These are the closest decision-layer precedents. MOSAIC's distinct statistical object is one confidence region for a sufficient table: it covers an uncountable channel family without testing channels separately or paying a channel-count correction.

Shift, contraction, and abstention. Dobrushin coefficients quantify channel contraction and have been used for privacy mechanisms and invariant domain generalization (Diaz, Sankar, and Kairouz 2018; Shui, Wang, and Gagné 2022). Fair-learning robustness and impossibility results show that unrestricted shift precludes nontrivial universal guarantees (Lechner et al. 2021; Agarwal et al. 2025; David et al. 2010). Benchmark and subgroup-robustness work documents the corresponding empirical failures (Koh et al. 2021; Sagawa et al. 2020; Zhou et al. 2021; Subbaswamy, Adams, and Saria 2021; Hashimoto et al. 2018; Sohoni et al. 2020; Kang et al. 2022; Ehyaei, Farnadi, and Samadi 2025). We therefore state the common-transform plus differential-contamination model explicitly and certify only conditionally on membership in it. Explicit abstention follows selective-prediction practice (El-Yaniv and Wiener 2010; Geifman and El-Yaniv 2017, 2019; Timans et al. 2025), but here it is triggered by a joint privacy–utility release contract rather than predictive confidence alone.

3 Problem Setup

Let $S \in [G]$ denote source, $Y \in [J]$ task label, and $C \in [K]$ the fine token. For a fixed label, write $p_s(c) = \Pr(C = c \mid S = s, Y = y)$. The source-blind release channel M produces $Z \in [L]$. The balanced accuracy of the strongest downstream source attacker is

$$A(P, M) = \frac{1}{G} \sum_{z=1}^{L} \max_{s \in [G]} (p_s M)(z), \tag{1}$$

with normalized advantage $\mathrm{Adv}_G(A) = (GA-1)/(G-1)$. This is chance-normalized and ranges from zero to one.

For each supported source–label stratum, the certification table supplies an empirical law $\widehat{p}_{s,y}$ and radius $\epsilon_{s,y}$. The simultaneous event

$$\mathcal{E} = \bigcap_{s,y} \{\|\widehat{p}_{s,y} - p_{s,y}\|_1 \le \epsilon_{s,y}\} \tag{2}$$

has probability at least $1 - \delta$ after a fixed allocation across strata. We use the Weissman multinomial radius (Weissman et al. 2003) in the confirmation, although any valid simultaneous region can replace it. The tokenizer and fine alphabet must be fixed without these observations.

The external law for label y belongs to the structured class

$$q_s = t_y p_s T_y + (1 - t_y) r_s, \qquad t_y \ge 1 - \eta_y, \tag{3}$$

where T_y is a source-independent fine-token channel from a registered convex uncertainty set and each residual r_s is arbitrary. Both components pass through the same release channel M. This placement is crucial: a constant-row software channel cannot leak information that appears before it.

4 Adaptive Universal-Attacker Certificate

For a bounded vector v, define the exact confidence-set support function

$$\Phi(\widehat{p}, \epsilon; v) = \max_{p \in \Delta_K : \|p - \widehat{p}\|_1 \le \epsilon} p^\top v. \tag{4}$$

It is obtained by moving at most $\epsilon/2$ mass from low to high coordinates of v, or from an equivalent linear-program dual. For attacker assignment $a \in [G]^L$, let $v_{M,a,s}(c) = \sum_z M(c, z) \mathbf{1}\{a(z) = s\}$. We then define

$$\overline{A}(\widehat{P}, M) = \frac{1}{G} \max_{a \in [G]^L} \sum_s \Phi(\widehat{p}_s, \epsilon_s; v_{M,a,s}). \tag{5}$$

Theorem 1 (Adaptive exact envelope)**.** *On $\mathcal{E}$, simultaneously for every row-stochastic M, including a channel selected as an arbitrary function of the same table, $A(P, M) \le \overline{A}(\widehat{P}, M)$. For fixed M, the right-hand side is the exact supremum over the product of the stated ℓ_1 confidence sets.*

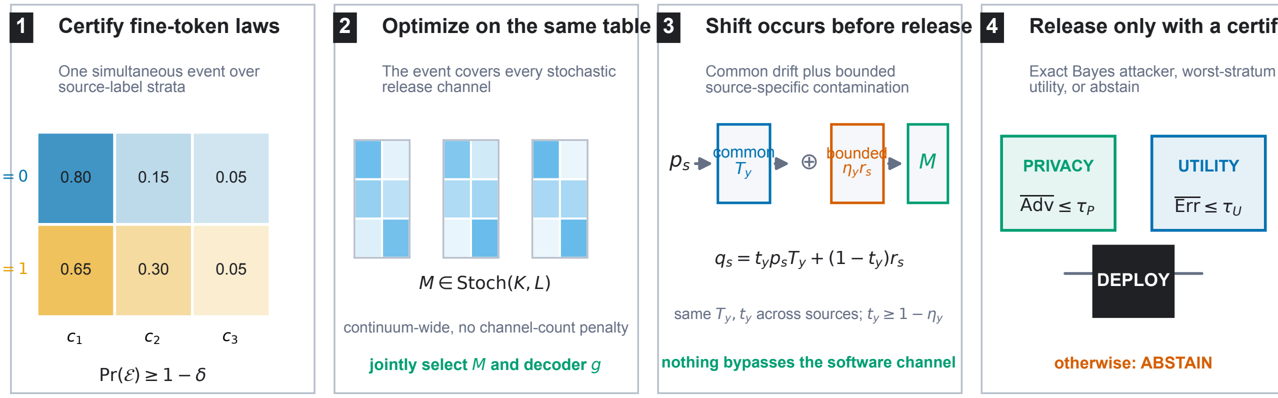

Figure 1: MOSAIC certifies fine-token laws before applying a source-blind stochastic release. The same confidence event covers continuum channel and decoder selection. Structured shift acts before the enforced channel, and a release occurs only when both universal-attacker privacy and worst-stratum task utility pass; otherwise MOSAIC abstains.

The proof writes the Bayes attacker as a maximum over deterministic assignments. For a fixed assignment the source confidence sets separate, giving one support function per source; all maxima are finite and commute. The implication is pointwise over the complete stochastic-channel space, which is why adaptive selection needs no channel-count correction.

5 Certification under Pre-Release Shift

Define the arbitrary differential-shift capacity

$$\mathrm{Cap}_G(M) = \frac{1}{G} \max_{a \in [G]^L} \sum_s \max_c v_{M,a,s}(c). \tag{6}$$

This equals $\sup_{r_1,\ldots,r_G} A(R, M)$. For $G = 2$, its normalized value is the Dobrushin coefficient, the largest total variation distance between two rows of M.

For a transform polytope with registered extremes $\mathcal{V}_y = \mathrm{ext}(\mathcal{T}_y)$, define the shifted envelope

$$\overline{A}_y^X(M) = \frac{1}{G} \max_{T \in \mathcal{V}_y,\, a \in [G]^L} \sum_s \Big[(1 - \eta_y)\Phi(\widehat{p}_{s,y}, \epsilon_{s,y}; T v_{M,a,s}) + \eta_y \max_c v_{M,a,s}(c) \Big]. \tag{7}$$

Theorem 2 (Transform-exact structured-shift certificate)**.** *On $\mathcal{E}$, every same-table selected M and every external law in Eq.* (3) *satisfy $A(Q, M) \leq \overline{A}_y^X(M)$. For fixed M, the right-hand side is the exact supremum over the stated confidence sets, transform polytope, retained masses, and arbitrary residual laws.*

For fixed T, retained mass, and attacker assignment, the source confidence sets and residual simplexes separate. Their support values are respectively $\Phi(\widehat{p}, \epsilon; T v)$ and $\max_c v(c)$. The latter is no smaller, so the worst retained mass is $1 - \eta_y$. The support objective is convex in T; its maximum over a polytope occurs at an extreme. These finite maxima commute, proving exactness. Since the statement holds pointwise for every M, it survives adaptive channel selection.

When transform extremes cannot be enumerated, a capacity-transfer corollary is available. Define $\rho_T(M) = \max_c \mathrm{TV}((TM)(c, \cdot), M(c, \cdot))$ and $B_y(M) = \min\{\mathrm{Cap}_G(M), \overline{A}_y(\widehat{P}, M) + \rho_{\mathcal{T}_y}(M)\}$. Then

$$A(Q, M) \leq (1 - \eta_y) B_y(M) + \eta_y \,\mathrm{Cap}_G(M). \tag{8}$$

The exact envelope in Eq. (7) is never larger. When $\eta_y = 0$ and $TM = M$, both reduce to the reference certificate; when $\eta_y = 1$, both reduce to exact distribution-free capacity.

Proposition 3 (Exactness value)**.** *For every M and confidence table, the transform-exact privacy envelope is no larger than Eq.* (8)*. The matching transform-exact utility value is likewise no larger than its capacity-transfer fallback.*

Both exact quantities are suprema over the registered structured class. The fallback proof upper-bounds every element of that same class using separate contraction and capacity inequalities, so taking the supremum preserves the ordering. The proposition holds pointwise in M; therefore optimizing the tighter envelope cannot worsen the globally optimal certified objective. It can still change feasibility sharply when a loose fallback straddles a contract threshold, which is the effect tested in Fig. 3.

For decoder $g : [L] \to [J]$, let $u_y(c) = \sum_z M(c, z)\mathbf{1}\{g(z) \neq y\}$. The matching exact conditional error certificate is

$$\overline{E}_{s,y}^X(M, g) = \max_{T \in \mathcal{V}_y} \Big[(1 - \eta_y) \Phi(\widehat{p}_{s,y}, \epsilon_{s,y}; T u_y) + \eta_y \max_c u_y(c) \Big]. \tag{9}$$

It is the exact supremum over the same confidence and shift sets. Thus M, g, privacy, and worst-stratum utility can all be selected on one event.

Theorem 4 (No free lunch)**.** *$\mathrm{Cap}_G(M) = 1/G$ if and only if all rows of M are identical. For two sources, if two fine tokens carry different task labels and incur row-wise decoder errors at most e_0 and e_1, then the normalized capacity is at least $1 - e_0 - e_1$.*

Consequently, unrestricted source-specific shift permits exact erasure only by making the public token independent of the fine token. A bounded η_y is therefore an identifiable assumption of the deployment contract, not a hidden technical convenience.

There is a separate identifiability boundary for external auditing. Suppose a required source is absent and let $A_{\text{miss}}(q_0, M) = \sup_{q_1 \in \Delta_K} A((q_0, q_1), M)$.

Theorem 5 (Missing-source audit)**.** *Any protocol with false-certification probability at most δ uniformly over the unobserved law q_1 can claim $A((q_0, q_1), M) \leq \tau < A_{\text{miss}}(q_0, M)$ with probability at most δ.*

The observed audit distribution is identical for every missing law; choosing a simplex maximizer makes every stronger claim a false certification. The same argument makes the utility bound for a missing source–label stratum at least $\max_c (M\ell_y)(c)$. Missing required strata are therefore unestimable, not evidence of safety; the supplement gives the full proof.

6 Global Optimization and Abstention

For fixed decoder, every support function in Eqs. (7) and (9), transformed score, and residual row maximum has a linear epigraph. Enforcing all registered transform extremes and all G^L attacker assignments gives one linear program in the KL channel entries. Enumerating all J^L decoders therefore yields the global finite-alphabet optimum. The capacity-transfer fallback uses binary branch variables and one mixed-integer linear program per decoder. A release is accepted only when the solver reports optimality, all primal residuals pass, and separately recomputed certificates agree with the objective; otherwise the rule abstains.

Release protocol. The end-to-end rule is deliberately mechanical. First, fix the tokenizer, transform polytope, source and task alphabets, thresholds, and confidence allocation without the certification fold. Second, form every supported $\widehat{p}_{s,y}$ and simultaneous radius. Third, for each of the J^L decoders, solve the exact LP minimizing worst certified utility subject to all privacy constraints. Fourth, replay every certificate from the returned channel and reject nonoptimal or numerically inconsistent solutions. Finally, among any separately registered tokenizers or erasers, apply their familywise allocation and select the feasible row with minimum certified error under a fixed tie break. No feasible row, unsupported required stratum, or failed solver check returns ABSTAIN; otherwise the software releases only (M, g) and the resulting token interface.

Let $R = \max_y |\mathcal{V}_y|$. For one decoder, a direct support-function formulation has order $KL + JRG^{L+1}K + JRGK$ variables and constraints: the middle term comes from privacy transform–attacker–source tuples and the last from utility strata. Thus the exact method is polynomial in the fine alphabet K for fixed (G, J, L, R) but exponential in the deliberately small released alphabet L. Decoder enumeration adds a factor J^L and parallelizes exactly. Larger public alphabets require separation or column-generation methods; the present theorem does not hide that computational boundary.

The exact optimizer is checked against binary confidence-ball endpoint enumeration, zero-radius population-risk identities, 100 randomized dominance comparisons with the capacity-transfer bound, and dense stochastic-channel grids. The fallback has separate randomized grid checks and a constructed witness where stochastic release strictly beats every deterministic channel. These checks validate the encodings but do not replace the algebra.

The locked analysis also predicts abstention as sample size changes. A finite-sample envelope follows by enlarging population-centered confidence balls: whenever the enlarged problem remains feasible, a Weissman union bound controls the probability that the empirical optimizer abstains. Because this bound can be loose near a decision boundary, we separately preregister a local delta-method power curve. Under a unique stable branch and strong regularity of the parametric linear program, centered simplex finite differences estimate the value gradient and the stratified multinomial central limit theorem predicts the deployment probability. A two-step gradient-stability check is required; the approximation is rejected if it fails.

7 Hash-Locked Evaluation

Synthetic confirmation. The synthetic confirmation fixes two source–label populations, two structured shift regimes, eight deployment rules, five possible per-stratum sample sizes, and 1,000 independent tables per registered cell. Pilot seeds are permanently excluded. Before confirmatory outcomes, the complete protocol, method code, theorem text, evaluators, pilot history, pass criteria, and theory predictions are SHA-256 locked and pushed to a public branch. Every selected channel is graded by a separately implemented exact population evaluator that enumerates common-transform extremes, attacker assignments, and residual simplex vertices. The confirmation stores every channel, decoder, seed, decision, confidence-event marker, and exact risk, then an independent script replays all rows and aggregate intervals. This artifact-first protocol follows established recommendations for model documentation, dataset documentation, internal auditing, and reproducible ML reporting (Mitchell et al. 2019; Gebru et al. 2021; Raji et al. 2020; Pineau et al. 2021).

The hard-safety regime fixes contamination $\eta = 0.20$, maximum source advantage $\tau_P = 0.30$, maximum worst-stratum error $\tau_U = 0.40$, and $n \in \{125, 250, 500\}$ observations per source–label stratum. The retention regime uses $(\eta, \tau_P, \tau_U) = (0.10, 0.35, 0.45)$ and $n \in \{125, 250, 500, 1000, 2000\}$. Both use familywise $\delta = 0.05$, three fine tokens, two released tokens, two registered common-transform extremes, and 1,000 untouched tables per cell. The eight rules are always-deploy, plug-in continuum selection, shift-unaware MOSAIC, held-out fixed-channel certification, finite-grid LTT, deterministic MOSAIC, MOSAIC, and an unattainable population oracle.

The primary safety contrast compares the plug-in continuum rule with MOSAIC at a near-boundary shift contract. The primary usefulness contrast compares safe retention against a held-out fixed-channel certificate, a finite-grid Learn-Then-Test rule with simultaneous one-sided bounds,

Rule	Deploy	False accept	Safe deploy
Hard safety boundary, $n = 125$			
Always deploy	100.0	57.6	42.4
Shift-unaware MOSAIC	80.3	76.6	3.7
Plug-in continuum	73.7	42.1	31.6
MOSAIC-transfer	0.0	**0.0**	0.0
Retention regime, $n = 250$			
Held-out fixed	2.5	0.0	2.5
Finite LTT	3.3	0.0	3.3
Deterministic MOSAIC	0.0	0.0	0.0
MOSAIC-transfer	**56.3**	0.0	**56.3**

Table 1: Primary registered rates (%). Each row uses the same 1,000 tables within a cell. “False accept” means deployed while violating at least one exact external privacy or utility contract.

		Utility threshold τ_U				
n	Certificate	.43	.44	.45	.46	.47
125	transfer	0.0	0.0	0.3	4.8	17.0
125	exact	11.6	29.0	53.0	76.9	94.4
250	transfer	4.6	23.6	57.8	88.0	98.3
250	exact	97.7	99.6	99.9	100	100

Table 2: Exploratory safe-deployment sensitivity (%) with fixed $\tau_P = .35$. Thresholds other than the locked .45 were examined after outcomes; all 20 cells have zero false acceptances.

and deterministic MOSAIC. The final gate additionally requires the preregistered theory curve to track deployment across sample sizes. All methods and cells remain reportable if any gate fails.

8 Locked Synthetic Results

All five preregistered gates passed. Here “MOSAIC-transfer” denotes the capacity-transfer fallback used before the transform-exact refinement. The independent replay recomputed 128,000 exact external risks and 128,000 deployment labels from 64,000 method-level receipts with zero mismatch. Figure 2 reports every registered deployment rule in the two primary cells; the population oracle is omitted from the bars because it is a feasibility diagnostic rather than an implementable method.

The safety contrast is paired: plug-in selection has 421 false acceptances that MOSAIC-transfer avoids and there are no reverse discordances. The paired rate difference is 42.1 points (bootstrap 95% interval $[39.0, 45.2]$; exact McNemar $p = 3.69\times10^{-127}$). Across all eight registered fallback cells, including 8,000 independently sampled certification tables, no accepted release violates the exact external contract. This is finite-run evidence consistent with the theorem, not a replacement for its $1-\delta$ guarantee.

In the retention cell, MOSAIC-transfer improves safe deployment over held-out certification by 53.8 points ($[50.6, 56.9]$), finite LTT by 53.0 points ($[49.9, 56.1]$), and deterministic MOSAIC by 56.3 points ($[53.2, 59.3]$); all paired exact p-values are below 2.1×10^{-158}. The stochastic comparison is essential: no deterministic channel satisfies the same finite-sample contract in this cell. Fallback deployment rises from 0.5% at $n = 125$ to 56.3% at $n = 250$, 99.8% at $n = 500$, and 100% at $n \geq 1000$. The preregistered local power approximation has 0.20-point mean absolute error over these five sizes and 0.44-point error at the primary size. Its active-set and two-step gradient-stability diagnostics pass at every registered point.

9 Transform-Exact Refinement

The broad comparison above uses the capacity-transfer certificate in Eq. (8), which remains useful when a transform family cannot be enumerated. We separately locked a paired confirmation of the exact envelope in Eq. (7): 1,000 untouched tables for one hard safety cell and four retention sample sizes, with both optimizers run on every table. The preregistered gates required pointwise dominance, zero failures on the simultaneous confidence event, false-acceptance rate at most δ, and large safe-retention margins at $n = 125$ and 250.

Figure 3 shows that exactness changes the release decision, not only the third decimal place of a bound. At $n = 125$, the exact method adds 527 safe deployments with no reverse discordance (paired exact $p = 4.55 \times 10^{-159}$). At $n = 250$, it adds 421 with no reverse discordance ($p = 3.69\times10^{-127}$). Mean certified worst error falls by 3.26 and 4.09 percentage points, respectively. The exact objective is no larger on all 5,000 paired tables and is strictly smaller on every retention table.

Both methods abstain on every table at the hard safety boundary: the exact envelope improves mean certified error by 0.45 points but correctly does not cross the contract. Across all ten method-by-cell combinations there are zero false acceptances; the per-cell exact 95% upper endpoint is 0.368%. The final strict-decision audit independently recomputes 20,000 privacy certificates, 40,000 utility certificates, 20,000 external risks, and all 10,000 decisions with zero mismatch. The public audit trail also retains an earlier failed decision-tolerance replay and the pre-rerun amendment that corrected it.

The post-hoc sweep in Table 2 checks whether the result is a single-threshold artifact. Transform-exact retention is higher at all ten sample-size–threshold combinations, including stricter contracts; this is descriptive sensitivity, not a second confirmatory test.

Official-method real-feature confirmation. A fresh paired pre-outcome lock evaluates five frozen feature stores spanning vision (Waterbirds), medical vision (Camelyon17-WILDS), text (CivilComments-WILDS and BiasBios), and gait time series (Sagawa et al. 2020; Koh et al. 2021; Borkan et al. 2019; De-Arteaga et al. 2019; Goldberger et al. 2000). For every dataset and each untouched seed in $\{105, \ldots, 109\}$, the construction split fits a balanced logistic task score and discretizes it at quartiles into four fine tokens. MOSAIC releases two tokens under identity or registered 10% adjacent-token smoothing, with $(\eta, \tau_P, \tau_U, \delta) = (0.05, 0.35, 0.49, 0.05)$. Each candidate is solved twice on the same table, once by the capacity-transfer fallback and once by the transform-exact certificate.

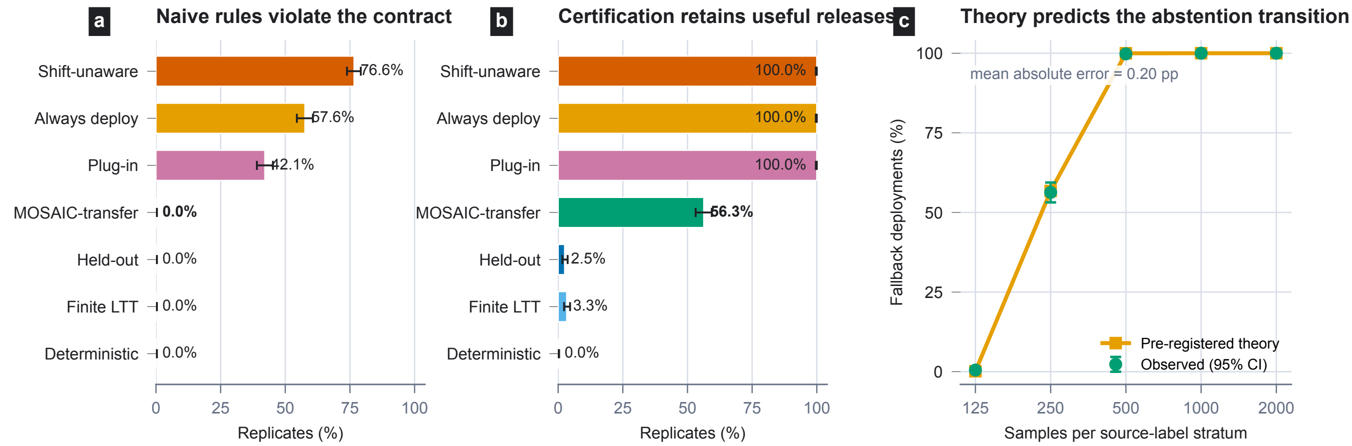

Figure 2: Hash-locked confirmation with 1,000 independent certification tables per cell. Error bars are exact 95% Clopper–Pearson intervals. (a) At the hard safety boundary, plug-in continuum selection falsely accepts 42.1% of tables, whereas MOSAIC-transfer abstains and records no false acceptance. Ignoring registered shift is worse. (b) In the feasible retention cell, MOSAIC-transfer safely deploys on 56.3% of tables, compared with 2.5% for a held-out fixed-channel certificate, 3.3% for finite LTT, and 0% for deterministic MOSAIC. Non-certifying rules happen to be safe in this easier cell but fail in panel (a). (c) The deployment transition follows the theory curve frozen before outcomes (mean absolute error 0.20 percentage points).

The frontier contains identity and twelve settings from the pinned official INLP, LEACE, R-LACE, TaCo, and MANCE++ implementations (Ravfogel et al. 2020; Belrose et al. 2023; Ravfogel et al. 2022a; Jourdan et al. 2023; Avitan, Goldberg, and Elazar 2026). Each edit has its own tokenizer, so the familywise event allocates $\delta/13$ per candidate; continuum channel selection within each fixed table incurs no additional correction. Among candidates that pass both certificates, the rule selects minimum certified worst-stratum error with a lexical tie break. Every receipt stores the complete token tables, channel, decoder, solver checks, decision, and natural external diagnostic. An independent program reconstructs every radius, certificate, selection, diagnostic, and all 650 global optima. External splits test empirical behavior but do not establish membership in Eq. (3); missing source–label strata are reported as unestimable rather than safe.

10 Official-Method Feature Results

All 25 dataset–seed jobs and all 325 registered candidate pairs completed with no proxy implementations or optimization failures. All ten preregistered gates pass. The independent replay reconstructed every confidence radius, certificate, selected channel, decoder, external diagnostic, lexical tie break, and 650 global optima with zero mismatch. Table 3 reports the complete selection behavior.

The paired comparison isolates the refinement. The exact objective is no larger on all 325 rows and is strictly smaller on 135, with mean and maximum reductions of 1.04 and 3.45 percentage points. Candidate deployments increase from 130 to 131. On Waterbirds seed 107, exact INLP rank 4 crosses the contract at certified error .465; its external error is .291 and source advantage is .057. This is the one additional selected deployment registered by the protocol, and it is safe on the natural diagnostic.

BiasBios supplies the repeated positive case. Every official candidate clears both exact contracts on all five seeds; MOSAIC selects INLP three times and R-LACE twice. The selected mean certified worst error is .291 (range .281–.297), while the natural diagnostic has mean worst error .207 and mean source advantage .047. All five selections are externally safe.

CivilComments and GaitPDB remain utility-limited: privacy is obtained only with certified error .50, so all ten jobs abstain. Four Waterbirds jobs also abstain. Camelyon17 yields five model-conditional releases, but every natural external split lacks a required source class; the missing-source theorem forces those diagnostics to remain unestimable. Across the six externally estimable exact selections there are zero false acceptances. This is a low-powered diagnostic (95% exact upper bound 45.9%), not evidence that the benchmark populations belong to Eq. (3).

11 Guarantee Semantics

Probability and deployment shift. The probability $1-\delta$ is over the certification table. On its simultaneous event, the conclusion holds uniformly over every optimized channel and every external distribution in the registered shift class. It does not assume that an external metric is within a validation confidence interval. Instead, $(\mathcal{T}_y, \eta_y)$ are explicit contract parameters: common transformations inside $\mathcal{T}_y$ and up to η_y arbitrary differential contamination are already inside the external supremum. A natural benchmark split can reveal a contradiction or missing support, but a favorable score cannot establish membership in that class.

Where multiplicity is paid. The tokenizer is frozen before certification, and one confidence event on its fine-token table covers the uncountable set of stochastic M and every decoder selected afterward. There is therefore no channel-count cor-

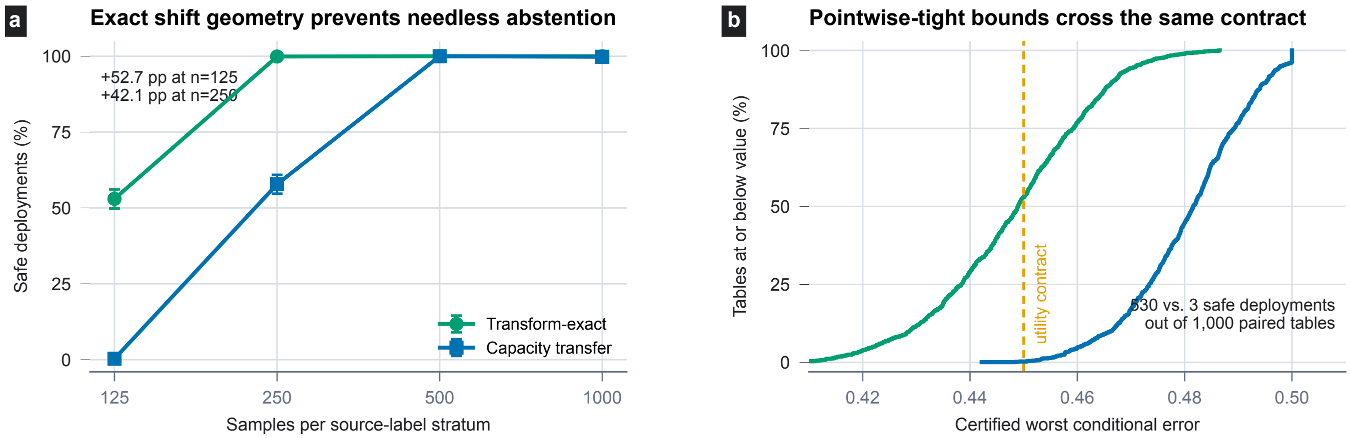

Figure 3: Audited paired transform-exact confirmation with 1,000 tables per point; error bars in (a) are exact 95% Clopper–Pearson intervals. The exact shift envelope safely deploys on 53.0% of low-data tables versus 0.3% for capacity transfer, and on 99.9% versus 57.8% at $n = 250$. Panel (b) shows the empirical CDF of the paired certified utility objectives at $n = 125$; exact geometry moves 527 additional tables across the same contract.

Dataset	Modality	Deploy F→X	Exact family	Exact error	External diagnostic
Waterbirds	vision	$0/5 \to 1/5$	INLP (1)	.490	5/65 rows safe; selected row safe
Camelyon17-WILDS	medical vision	$5/5 \to 5/5$	INLP (2), LEACE (3)	.248	unestimable: source absent
CivilComments-WILDS	text	$0/5 \to 0/5$	–	.500	0/65 candidate rows safe
BiasBios-Clinical	clinical text	$5/5 \to 5/5$	INLP (3), R-LACE (2)	.291	5/5 selected; 65/65 rows safe
GaitPDB	time series	$0/5 \to 0/5$	–	.500	0/65 candidate rows safe

Table 3: Fresh locked paired frontier, five seeds per dataset and 13 candidates per seed. F and X denote fallback and transform-exact certificates. “Best exact error” is the mean minimum exact value across the five jobs. External diagnostics are descriptive and do not establish structured-shift membership.

rection. If the frontier contains m separately learned erasers or tokenizers, their m different tables do require a registered familywise allocation, as in the official-method study. This is the precise distinction from applying LTT to a finite list of already constructed channels: MOSAIC pays for statistical objects, not for every point subsequently optimized inside one covered object.

Threat surface. The attacker knows M and may use any function of the released token Z; Eq. (1) is its exact balanced Bayes success, with no probe or computational restriction. The guarantee requires M to be source-blind and software-enforced and Z to be the only public representation interface. Releasing C, the original feature vector, source-dependent metadata, or a side channel falls outside the theorem. The task decoder may be public because utility is certified jointly rather than treated as secret.

12 Limitations and Scope

The guarantee is conditional on Eq. (3); an ordinary benchmark split cannot prove that a real hospital, text domain, or sensor environment belongs to that class. Membership requires an engineering specification or separate bridge data containing every required source–label stratum, with its own confidence allocation. Without that evidence, a favorable score cannot establish membership. The finite-token theorem also fixes the tokenizer and alphabet independently of the certification table; large alphabets may require structured confidence regions or column generation.

MOSAIC certifies source inference from the released token, not every possible harm associated with representation use. It does not protect information that is released through a separate interface, and it cannot recover utility that the privacy contract makes information-theoretically impossible. Finally, the novelty claim concerns the complete adaptive and shift-aware construction; its classical ingredients must remain attributed to their respective literatures.

13 Conclusion

MOSAIC treats representation editing as release-mechanism design rather than a probe contest. By certifying fine-token laws before a stochastic post-channel, it can optimize over a continuum on the same table while retaining a single finite-sample event. The structured shift theorem separates common drift that the channel can absorb from differential drift that its capacity must pay for, and the no-free-lunch result states when abstention is unavoidable. The locked evaluation shows both sides of that decision: abstention at an infeasible safety boundary and substantially higher safe retention when exact shift geometry can replace a generic capacity bound.

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